

Code Layout, Readability and Reusability

Date	01 July 2026
Team ID	
Project Name	RISING WATERS
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Consistent indentation followed throughout the project
2	Proper File Structure	Files and folders are logically organised	Yes	Project files are logically organized into folders
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Meaningful variable names are Used
4	Function / Method Names	Functions are descriptively named	Yes	Functions are named clearly based on their performance

7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code is minimized
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Errors are handled using exception handling
5	Code Comments	Inline and block comments explain logic	Yes	Comments are added to improve Code understanding
6	Modular Design	Code is split into reusable functions Or modules	Yes	Code is divided into reusable modules

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Flask Application	Python, Flask	Web interface and prediction request handling	High
2	Data Preprocessing Module	Python, Pandas, Numpy	Dataset cleaning and preprocessing	High
3	Machine Learning Model	Scikit-learn, XGBoost	Flood prediction engine	High
4	Prediction Function	Python	Predicting flood risk from User inputs	Medium
5	HTML Templates	HTML, CSS	User interface pages	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Project structure is clean, organized, and easy to navigate
Readability	4	Meaningful variable names and comments improve readability
Reusability	5	Modules can be reused in future flood prediction projects
Documentation / Comments	4	Adequate comments and documentation are provided but comments can be expanded further
Overall Score	4.5	The project follows good coding practices with a modular architecture, good readability